13/06/2026

CODE-P

Aakash
Medical | IIT-JEE | Foundations

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PROJECT PHOENIX FOR RE-NEET-2026

CST-09

MM: 720 **Time: 180 Mins.**

Answers

1. (4)	37. (3)	73. (4)	109. (2)	145. (4)
2. (3)	38. (3)	74. (2)	110. (1)	146. (3)
3. (4)	39. (2)	75. (2)	111. (4)	147. (3)
4. (3)	40. (3)	76. (4)	112. (2)	148. (4)
5. (3)	41. (3)	77. (1)	113. (1)	149. (2)
6. (2)	42. (2)	78. (4)	114. (4)	150. (2)
7. (4)	43. (1)	79. (2)	115. (3)	151. (3)
8. (2)	44. (4)	80. (3)	116. (2)	152. (4)
9. (3)	45. (4)	81. (1)	117. (3)	153. (4)
10. (4)	46. (2)	82. (3)	118. (4)	154. (3)
11. (3)	47. (4)	83. (4)	119. (3)	155. (4)
12. (4)	48. (1)	84. (3)	120. (4)	156. (1)
13. (2)	49. (4)	85. (3)	121. (2)	157. (1)
14. (4)	50. (1)	86. (1)	122. (1)	158. (2)
15. (3)	51. (4)	87. (3)	123. (1)	159. (4)
16. (1)	52. (3)	88. (2)	124. (1)	160. (4)
17. (3)	53. (2)	89. (3)	125. (2)	161. (4)
18. (4)	54. (1)	90. (2)	126. (4)	162. (2)
19. (3)	55. (3)	91. (2)	127. (2)	163. (3)
20. (2)	56. (4)	92. (2)	128. (4)	164. (1)
21. (3)	57. (4)	93. (2)	129. (4)	165. (3)
22. (3)	58. (2)	94. (4)	130. (3)	166. (4)
23. (1)	59. (2)	95. (4)	131. (3)	167. (4)
24. (4)	60. (2)	96. (3)	132. (4)	168. (3)
25. (3)	61. (1)	97. (2)	133. (1)	169. (2)
26. (2)	62. (4)	98. (3)	134. (3)	170. (4)
27. (3)	63. (2)	99. (1)	135. (2)	171. (4)
28. (4)	64. (2)	100. (3)	136. (3)	172. (1)
29. (4)	65. (1)	101. (3)	137. (2)	173. (1)
30. (4)	66. (4)	102. (3)	138. (3)	174. (3)
31. (1)	67. (1)	103. (2)	139. (3)	175. (2)
32. (1)	68. (2)	104. (3)	140. (4)	176. (4)
33. (3)	69. (3)	105. (3)	141. (2)	177. (3)
34. (4)	70. (4)	106. (2)	142. (2)	178. (3)
35. (3)	71. (2)	107. (2)	143. (3)	179. (3)
36. (3)	72. (1)	108. (1)	144. (1)	180. (3)

Hints and Solutions

PHYSICS

(1) **Answer :** (4)
Solution:
Two physical quantity having same dimensions may have different unit like torque and energy.
A dimensionless quantity may have unit like plane angle has unit radian but it is dimensionless.

(2) **Answer :** (3)
Solution:
$v = t - 4$
$\Rightarrow s = \int_0^4 \left(4 - t\right) dt + \int_4^8 \left(t - 4\right) dt$
$\Rightarrow s = 8 + 8 = 16$ m
$\Rightarrow v_{avg} = \frac{16}{8} = 2$ m/s

(3) **Answer :** (4)
Solution:
Let initial speed of particle is 'u' & its acceleration is 'a'.
$S_n = u + \frac{a}{2}\left[2n - 1\right]$
$10 = u + \frac{a}{2}(9)$ …(i)
$v = u + at$
$20 = u + 5a$ …(ii)
By subtracting equation (i) from (ii)
$10 = 5a - 4.5a$
$a = 20$ m/s^2

(4) **Answer :** (3)
Solution:
$\omega^2 R = 4t^2$
$\omega = \sqrt{\frac{4}{R}\, t^2} = \sqrt{\frac{4}{R}}\, t$
$\therefore \ \frac{d\omega}{dt} = 1$
$\therefore \alpha = 1$ rad/s^2
$\theta = \frac{1}{2}\alpha t^2$
$\Rightarrow \theta = 2$ rad

(5) **Answer :** (3)
Solution:
$V_m = \frac{320}{40} = 8 \text{ m/s}$
$V_R = 8 \times \frac{3}{4} = 6 \text{ m/s}$
$d = \sqrt{(320)^2 + (40 \times 6)^2} = 400 \text{ m}$

(6) **Answer :** (2)
Solution:
Height attained after nth rebound is
$h_n = e^{2n} h$

Distance travelled till 3rd collision is
$s = h + 2 \times e^2 h + 2 \times e^4 h$
$s = 64 + 2 \times \left(\frac{1}{2}\right)^2 \times 64 + 2 \times \left(\frac{1}{2}\right)^4 \times 64$
$s = 104 \text{ m}$

(7) **Answer :** (4)
Solution:

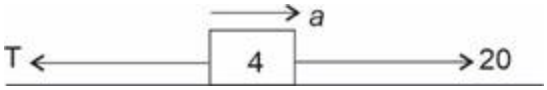
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$P = \frac{W}{t} = \frac{\frac{1}{2}mv^2}{t} \Rightarrow v \propto t^{1/2}\ (P = \text{constant})$

$s = \int_0^t v\,dt = \int_0^t t^{1/2}\,dt = \frac{2t^{3/2}}{3}$

$s \propto t^{3/2}$

(8) **Answer :** (2)
Solution:



Common acceleration = $\frac{20-10}{(2+4)} = \frac{10}{6}$ m s^{-2}

By FBD
$20 - T = 4 \times a$
$T = 20 - 4a = 20 - \frac{4\times10}{6} = \frac{80}{6} = \frac{40}{3}$ N

(9) **Answer :** (3)
Solution:
$0 < f_s \leq f_\ell$

(10) **Answer :** (4)
Solution:
$W = U_f - U_i$

$W = -3Gm^2/2a - (-3Gm^2/a)$

$W = 3Gm^2/2a$

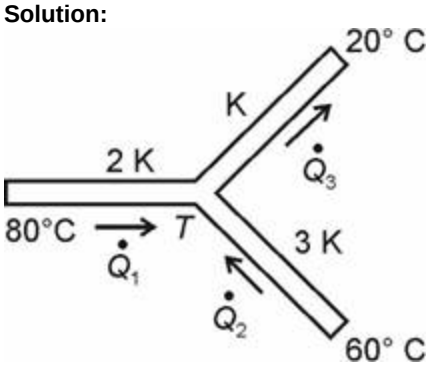
(11) **Answer :** (3)
Solution:
MOI depends on the choice of axis of rotation also.

(12) **Answer :** (4)
Solution:

If $\vec{F}_{ext} = 0$, then velocity of centre of mass will remain same.

Hence velocity of centre of mass = 0

(13) **Answer :** (2)
Solution:



Take junction temperature to be T at steady state.

$\dot{Q}_1 + \dot{Q}_2 = \dot{Q}_3$

$\Rightarrow \frac{2kA}{l}(80 - T) + \frac{3kA}{l}(60 - T) = \frac{kA}{l}(T - 20)$

$\Rightarrow 160 - 2T + 180 - 3T = T - 20$
$\Rightarrow 6T = 360°$
$T = 60°C$

(14) **Answer :** (4)
Solution:

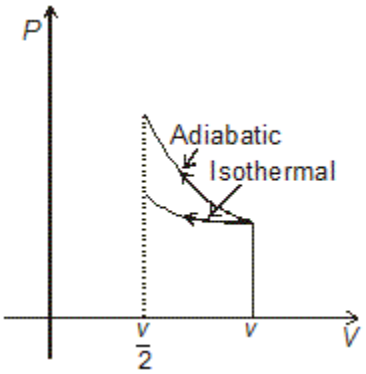
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Rate of outflow $= \frac{\text{volume of liquid}}{\text{time}}$

$= Av = \pi r^2 \times \sqrt{2gh}$

(15) **Answer :** (3)
Solution:
Young modulus for solids only (A – *R*)
Shear modulus for solids only (*B* – *R*)
Bulk modulus for both solids and fluids (*C* – *Q*)

(16) **Answer :** (1)
Solution:



(17) **Answer :** (3)
Solution:
P = *mV* (*PV* = *nRT*)

$\frac{nRT}{V} = mV$

$T = \frac{m}{nR}V^2 \Rightarrow T \propto V^2 \Rightarrow \frac{T_A}{T_B} = \left(\frac{V_A}{V_B}\right)^2$

$\frac{300}{T_B} = \left(\frac{V_0}{2V_0}\right)^2$

$T_B = 4 \times 300$ = 1200 K
= 927°C

(18) **Answer :** (4)
Solution:

$U_{initial} = 5\left(\frac{1}{2}nRT\right)$

$U_{final} = 3\left(\frac{1}{2}(2n)RT\right)$

$\mathrm{R}_{eq} = U_{final} - U_{ini}$

$= \frac{nRT}{2}$

(19) **Answer :** (3)
Solution:
Frequency of input and output remains unchanged in transformer.

(20) **Answer :** (2)
Solution:

$V_0 = A\omega, x = \frac{A}{2}$

$V = \omega\sqrt{A^2 - x^2} = \omega\sqrt{A^2 - \frac{A^2}{4}} \Rightarrow V = \omega\sqrt{\frac{3}{4}A^2}$

$\frac{\sqrt{3}}{2}A\omega = \frac{\sqrt{3}}{2}V_0$ or $x = A\sin\omega t = \frac{A}{2}$

$\sin\omega t = \frac{1}{2} \Rightarrow \cos\omega t = \frac{\sqrt{3}}{2}$

$V_1 = V_0 = \frac{V_0\sqrt{3}}{2}$

(21) **Answer :** (3)
Solution:
Hint : $\Delta\phi = \frac{2\pi}{\lambda}\Delta x$

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Sol. : $v = f\lambda \Rightarrow \lambda = \frac{v}{f} = \frac{300}{600} = \frac{1}{2}$ m

Now, $\Delta\phi = \frac{2\pi}{\lambda}\Delta x \Rightarrow \Delta x = \frac{\lambda \cdot \Delta\phi}{2\pi} = \frac{1}{2} \times \frac{\pi}{6 \times 2\pi}$

$= \frac{1}{24}\text{m} = 0.041$ m

(22) **Answer :** (3)
Solution:
Let $x = A\sin(\omega t + \phi)$
$\therefore v = A\omega\cos(\omega t + \phi)$
Given that at $t = 0$, $v = -A\omega$
$-A\omega = -A\omega\cos\phi$
$\Rightarrow \cos\phi = -1$
$\therefore \phi = \pi$
$\therefore$ Equation of SHM is $x = A\sin(\omega t + \pi)$

(23) **Answer :** (1)
Solution:
Displacement wave and pressure wave have phase difference equal to $\frac{\pi}{2}$ radians.

(24) **Answer :** (4)
Solution:
Total charge on 3μF before connecting = 300 μC
Charge on 3 μF after distribution = 60 μC
So amount of charge flow = 240 μC

(25) **Answer :** (3)
Solution:
$E = pc = 2 \times 10^{-11} \times 3 \times 10^{10} = 6 \times 10^{-1}$ erg
$= 6 \times 10^{-1} \times 10^{-7}$ J $= 6 \times 10^{-8}$ J

(26) **Answer :** (2)
Solution:
Hint: On equatorial line, potential = 0

Sol.:

$E_1 = 0 \ \& \ E_2 \neq 0$

$\Rightarrow \quad E_1 + E_2 = E_2$

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(27) **Answer :** (3)
Solution:
$U = -PE\cos\theta$
For stable equilibrium, potential energy will be minimum.
$U = U_{min}$
$\cos\theta = 1$
$\Rightarrow \theta = 0°$

(28) **Answer :** (4)
Solution:
Charge of an isolated system is conserved.
Also $q = ne$

(29) **Answer :** (4)
Solution:
$F = ILB \sin\theta$
$\Rightarrow \frac{F}{L} = IB\sin\theta \ = 2 \times 10^{-3} \times 4 \times \frac{1}{2} = 4 \times 10^{-3} = 4 \ \text{mN m}^{-1}$

(30) **Answer :** (4)
Solution:
Susceptibility of a diamagnetic substance is independent of temperature.

(31) **Answer :** (1)
Solution:

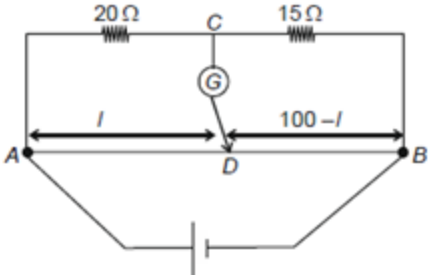
Stopping potential depends on frequency of the incident light.

(32) **Answer :** (1)
Solution:
$R_{eq} = 4\ \Omega$

$I = \frac{V}{R_{eq}} = \frac{8\ V}{4\ \Omega}$

$I = 2\ A$

(33) **Answer :** (3)
Solution:



Since $\frac{R_1}{R_2} = \frac{l}{100-l}$

$\Rightarrow \quad \frac{20}{15} = \frac{l}{100-l}$

$\Rightarrow$ 400 – 4l = 3l

$\Rightarrow \quad l = \frac{400}{7}\ \text{cm}$

$\boxed{l = \frac{4}{7}\ m\ \text{ from }\ A}$

(34) **Answer :** (4)
Solution:
All the inductors are shorted.

(35) **Answer :** (3)
Solution:
For a thin prism
$\delta = (\mu - 1)A = (\sqrt{2} - 1)4^\circ = 1.656^\circ$

(36) **Answer :** (3)
Solution:
Hint:

$I = \frac{1}{2}\varepsilon_0 E_0^2 c = \frac{B_0^2 c}{2\mu_0}$

Sol.: since,
$c = \frac{1}{\sqrt{\mu_0 \varepsilon_0}}$

$\therefore \quad I = \frac{1}{2}\varepsilon_0 E_0^2 \cdot \frac{1}{\sqrt{\mu_0 \varepsilon_0}} = \frac{1}{2}\sqrt{\frac{\varepsilon_0}{\mu_0}} E_0^2$

(37) **Answer :** (3)
Solution:

$y_{(1)\ \text{min}} = \frac{\lambda_1 D}{d}\ y_{(n)\ \text{max}} = \frac{\left(n + \frac{1}{2}\right)\lambda_2 D}{d}$

$\frac{\lambda_1 D}{d} = \frac{\left(3 + \frac{1}{2}\right)\lambda_2 D}{d}$

$\lambda_1 = \frac{7}{2}\lambda_2 = 3.5\ \lambda_2$

(38) **Answer :** (3)
Solution:

Hint: Use $I = I_0 \cos^2\theta$

Sol.: $I = I_0 \cos^2\theta$

$\frac{I_0}{2} = I_0 \cos^2\theta$

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$\therefore \cos\theta = \frac{1}{\sqrt{2}}$

θ = 45°

(39) **Answer :** (2)
Solution:
f = – 12 cm *o* = 4 cm *I* = 1 cm

$m = -\frac{v}{v} = \frac{1}{0} = \frac{-1}{4}$

$v = \frac{u}{4}$

$\frac{1}{-12} = \frac{1}{\frac{u}{4}} + \frac{1}{t} = \frac{4}{t} + \frac{1}{t} = \frac{5}{t}$

⇒ *u* = – 60 cm

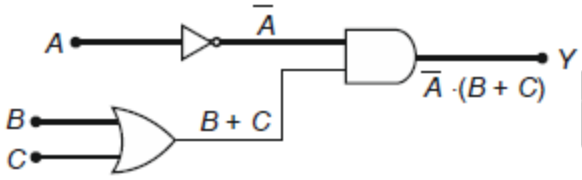
(40) **Answer :** (3)
Solution:
Energy incident $= \frac{hc}{\lambda} = \frac{6.63 \times 10^{-34} \times 3 \times 10^{8}}{975 \times 10^{-10} \times 1.6 \times 10^{-19}}$ eV
= 12.75 eV
The Hydrogen atom will be excited to *n* = 4
Number of spectral lines $= \frac{4(4-1)}{2} = 6$

(41) **Answer :** (3)
Solution:
In both (Positive and negative) half cycle, two diodes will be ON in each case and output will be equal to

$V_0 = \begin{cases} V_i\,; & +ve\ cycle \\ -V_i\,; & -ve\ cycle \end{cases}$

(42) **Answer :** (2)
Solution:
$V = \frac{h}{e}(v - v_0)$

(43) **Answer :** (1)
Solution:



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(44) **Answer :** (4)
Solution:
$\frac{\Delta r}{r} \times 100 = \frac{0.005}{0.25} \times 100 = \frac{0.5}{0.25} = 2\%$

(45) **Answer :** (4)
Solution:
Due to heating, number of electron-hole pairs will increase, so overall resistance of diode will change.

CHEMISTRY

(46) **Answer :** (2)
Solution:
In case of positive deviation, A – B interactions are weaker than those between A – A or B – B interactions.

(47) **Answer :** (4)
Solution:
$\Delta T_f = K_f \times m$ ⇒ 0.93 = $0.93 = 1.86 \times \frac{36}{M} \times \frac{1}{1.2}$
M = 60 and 30 × n = 60 ∴ n = 2
So compounds is $C_2H_4O_2$.

(48) **Answer :** (1)

Hint:

Degree of dissociation $(\alpha) = \frac{\Lambda_m}{\Lambda_m^\circ}$

Solution:

$\alpha = \frac{12}{400} = 0.03$

For weak acid, K = $C\alpha^2$

K = $0.05 \times (0.03)^2$

K = 4.5×10^{-5}

(49) Answer : (4)

Hint:

Rate of reaction ∝ $[A]^x [B]^y [C]^z$

Solution:

Let rate law expression is

Rate = K $[A]^x [B]^y [C]^z$

Case-I : $(2)^{x+y} = 2 \Rightarrow x + y = 1$

Case II : $(2)^{y+z} = 4 = (2)^2 \Rightarrow y + z = 2$

If x = 0, then y = z = 1

∴ The possible rate law expression is

Rate = K $[A]^0 [B]^1 [C]^1$

(50) Answer : (1)

Solution:

$4OH^- \rightarrow 2H_2O + O_2 + 4e^-$

for 2 mole of H_2O = 4F charge is required

for 1 mole of H_2O = $\frac{4F}{2}$ = 2F required

$\overset{+7}{M}nO_4^- \rightarrow \overset{+2}{M}n^{2+}$

for 1 mole MnO_4^- = 5F charge is required

$Ca^{2+} \xrightarrow{+2e^-} Ca$

for 1 mole Ca^{2+} ion required = 2F

1.5 mole Ca^{2+} ion required = $\frac{2}{1} \times 1.5$ = 3F

$\overset{+2}{F}eO \rightarrow \overset{+3}{F}e_2O_3$

for 1 mole FeO, 1F charge is required.

(51) Answer : (4)

Hint:

According to collision theory of chemical reactions,

Rate of reaction (Rate) = $Z_{AB} e^{-E_a/RT}$

Solution:

⇒ For bimolecular elementary reaction,

A + B → Products

Rate of reaction each be expressed as:

Rate = $Z_{AB} e^{-E_a/RT}$

AB represents the collision frequency of reactants, A and B.

⇒ Catalyst catalyses the spontaneous reaction but does not catalyse non-spontaneous reactions.

⇒ $\underset{\text{(Cane sugar)}}{C_{12}H_{22}O_{11}} + H_2O \xrightarrow{H^+} \underset{\text{(Glucose)}}{C_6H_{12}O_6} + \underset{\text{(Fructose)}}{C_6H_{12}O_6}$

Rate = k[$C_{12}H_{22}O_{11}$]

(52) Answer : (3)

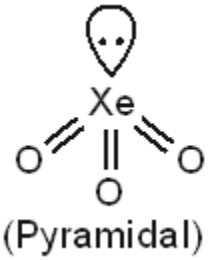
Solution:

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- $NH_4NO_3 \xrightarrow{\Delta} N_2O + 2H_2O$
- $2NaNO_2 + 2FeSO_4 + 3H_2SO_4 \rightarrow Fe_2(SO_4)_3 + 2NaHSO_4 + 2H_2O + 2NO$
- $2Pb(NO_3)_2 \xrightarrow{673K} 4NO_2 + 2PbO + O_2$
- $4HNO_3 + P_4O_{10} \rightarrow 4HPO_3 + 2N_2O_5$

(53) **Answer :** (2)
Solution:
Chlorine – Deacon's process
Ammonia – Haber's process
Sulphuric acid – Contact process
Nitric acid – Ostwald process

(54) **Answer :** (1)
Hint:
$XeF_6 + 3H_2O \rightarrow XeO_3 + 6HF$
Solution:



(Pyramidal)

(55) **Answer :** (3)
Hint:
Tb : $[Xe]4f^9\,6s^2$

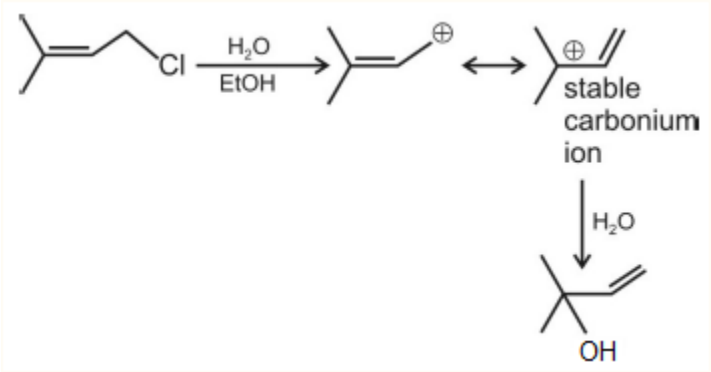
(56) **Answer :** (4)
Solution:
Correct order of enthalpy of atomization :
V > Ni > Co > Cu

(57) **Answer :** (4)
Hint:
Ionisation isomerism arises when the counter ion in a complex salt is itself a potential ligand and can displace a ligand which can then become the counter ion.
Solution:

Complex	**Magnetic character**	**Isomerism**
$[Cr(NH_3)_4Br_2]Br$	Cr^{3+} is d^3, hence paramagnetic	Cis-trans
$[Ti(H_2O)_5Br](NO_3)_2$	Ti^{3+} is d^1, hence paramagnetic	Ionization
$[Pt(en)(NH_3)Br]NO_3$	Pt^{2+} is d^8, complex is square planar, all electrons are paired	Ionization
$[Co(NH_3)_4(NO_3)_2]NO_3$	Co^{3+} is d^6, hence diamagnetic	Cis-trans

(58) **Answer :** (2)
Hint:
More stable the carbonium ion, faster will be the solvolysis.
Solution:

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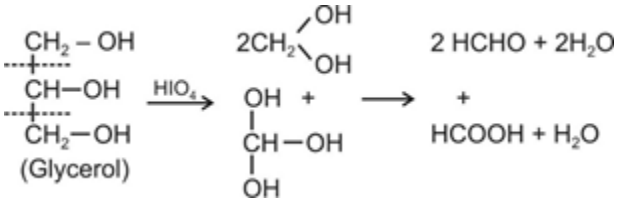


(59) **Answer :** (2)

Hint:

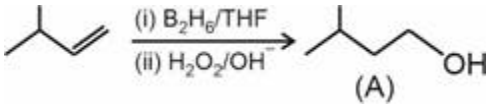
Cleavage of C–C bond take place.

Solution:

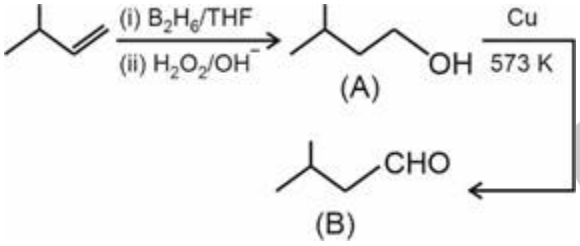


(60) **Answer :** (2)

Hint:



Solution:



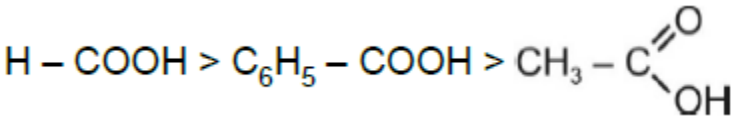
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(61) **Answer :** (1)

Hint:

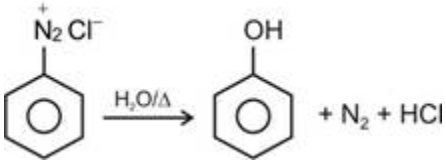
+I effect causing group decreases the acidic strength of carboxylic acid.

Solution:



(62) **Answer :** (4)

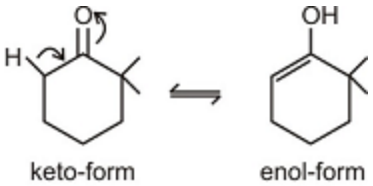
Solution:



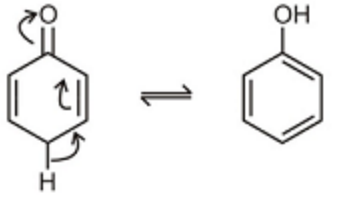
(63) **Answer :** (2)

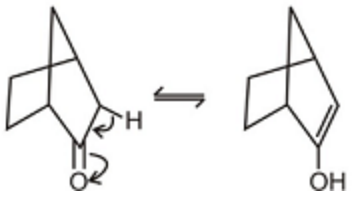
Solution:

The compounds having enolisable proton will show keto-enol tautomerism.

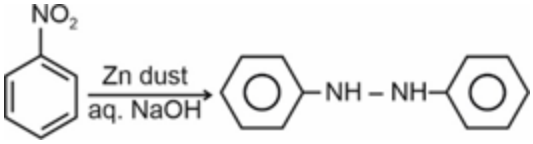


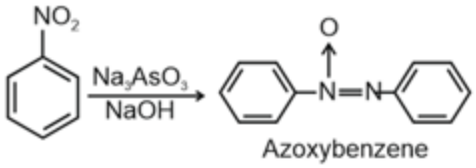
keto-form enol-form



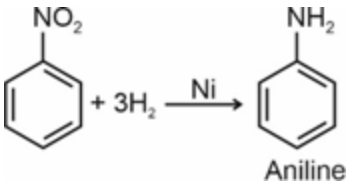


(64) Answer : (2)
Solution:

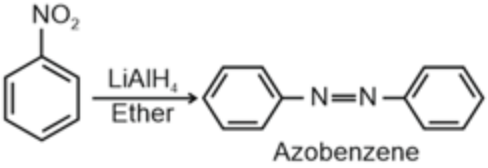




Azoxybenzene



Aniline



Azobenzene

(65) Answer : (1)
Solution:

Nucleic Acid	Sugar
DNA RNA	β-D-2 deoxyribose sugar β-D-ribose

(66) Answer : (4)
Solution:
3° Amines do not react with Hinsberg's reagent.

(67) Answer : (1)
Solution:
Amylose is water soluble component of starch. Chemically amylose is a long unbranched chain with 200 – 1000 α-D-(+)-glucose units held together by C1 – C4 glycosidic linkage.

(68) Answer : (2)

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Solution:

Number of glucose molecules = $200 \times 0.4 \times 10^{-3} \times 6.02 \times 10^{23} = 481.6 \times 10^{20} = 4.8 \times 10^{22}$

(69) **Answer :** (3)
Solution:

S(z = 16) : $1s^2 2s^2 2p^6 3s^2 3p^4$
Total number of p-subshells = 2*p* & 3*p*
Total number of electrons in p-subshells = 6 + 4 = 10

(70) **Answer :** (4)
Hint:
In equal masses of O_2 and N_2, N_2 will have more number of atoms.
Solution:
Molality and mole fraction are temperature independent quantities.
Mole of $O_2 = \frac{w}{32}$
Number of atoms in O_2 = $\frac{w}{32} \times N_A \times 2 = \frac{w}{16} N_A$
Number of atoms in N_2 = $\frac{w}{28} \times N_A \times 2$
$= \frac{w}{14} \times N_A$

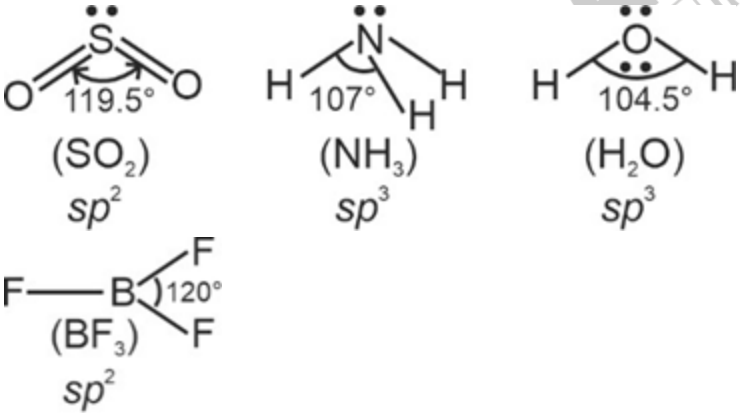
(71) **Answer :** (2)
Solution:

n	*l*	Orbital
2	1	2*p*
3	2	3*d*
1	0	1*s*
3	1	3*p*

(72) **Answer :** (1)
Hint:
Element X is chlorine
Solution:

Chlorine belongs to halogen family *i.e.* group number 17th and period number 3.

(73) **Answer :** (4)
Solution:



(74) **Answer :** (2)
Hint:
If number of bonding and antibonding electrons are equal then species does not exist.
Solution:
Be_2 has equal bonding and antibonding electrons, so does not exist.

(75) **Answer :** (2)
Hint:
$\Delta H = \Delta U$ if $\Delta n_g = 0$
Solution:
$C(s) + O_2(g) \rightarrow CO_2(g)$
$\because$ $\Delta ng = nP - nR = 1 - 1 = 0$

$\therefore \quad \Delta H = \Delta U$

(76) **Answer :** (4)
Solution:
$N_2 + 3H_2 \rightarrow 2NH_3$
ΔH for the formation of 2 mole of NH_3 = (x + 3y – 6z) kJ

(77) **Answer :** (1)
Hint:
Common ion effect

$$M(OH)_2 + H_2O(l) \rightleftharpoons M^{2+}(aq) + 2OH^-(aq)$$

At initial	0	0.2
At equilibrium state	s	$0.2+2s \approx 0.2$

$K_{sp} = [M^{2+}] [OH^-]^2$
$4.0 \times 10^{-15} = (s) (0.2)^2$

$s = \frac{4\times10^{-15}}{4\times10^{-2}}$
$= 10^{-13}$ mol/L

(78) **Answer :** (4)
Hint:
On adding equations, equilibrium constants get multiplied.
Solution:

A $\rightleftharpoons$ B; $K_C = 10$

B $\rightleftharpoons$ C; $K_C = 5$

C $\rightleftharpoons$ D; $K_C = 2$

A $\rightleftharpoons$ D; $K_C = 10 \times 5 \times 2 = 100$

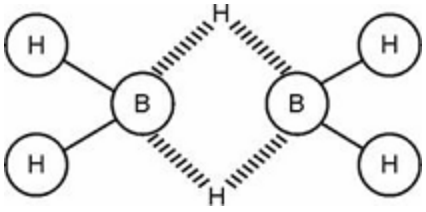
(79) **Answer :** (2)
Hint:
Redox reactions are those in which both oxidation and reduction takes place.
Solution:
• Combination redox reaction:

$\mathrm{CH_4(g) + 2O_2(g) \xrightarrow{\Delta} CO_2(g) + 2H_2O(l)}$
• Decomposition redox solution:

$\mathrm{2KClO_3(s) \xrightarrow{\Delta} 2KCl(s) + 3O_2(g)}$
• Metal displacement redox reaction:

$\mathrm{V_2O_5(s) + 5Ca(s) \xrightarrow{\Delta} 2V(s) + 5CaO(s)}$
• Non-metal displacement redox reaction:
$\mathrm{Fe(s) + 2HCl(aq) \rightarrow FeCl_2(aq) + H_2(g)}$

(80) **Answer :** (3)
Hint:
Hybridisation of B atoms in B_2H_6 is sp^3.
Solution:



Diborane has 2 three centre two electron bonds.

(81) Answer : (1)
Solution:

$$\%\ \text{of Br} = \frac{80}{188} \times \frac{\text{mass of AgBr} \times 100}{\text{mass of compound}}$$

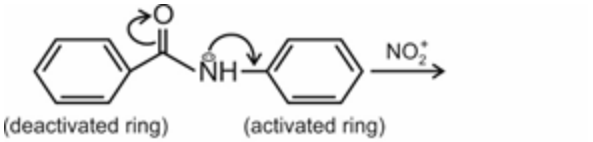
$\because$ 188 g AgBr contains 80 g bromine

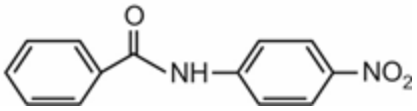
$\therefore$ 0.15 g AgBr contains $\frac{80 \times 0.15}{188}$ g bromine

so % bromine = $\frac{80 \times 0.15 \times 100}{188 \times 0.2}$ = 31.9 %

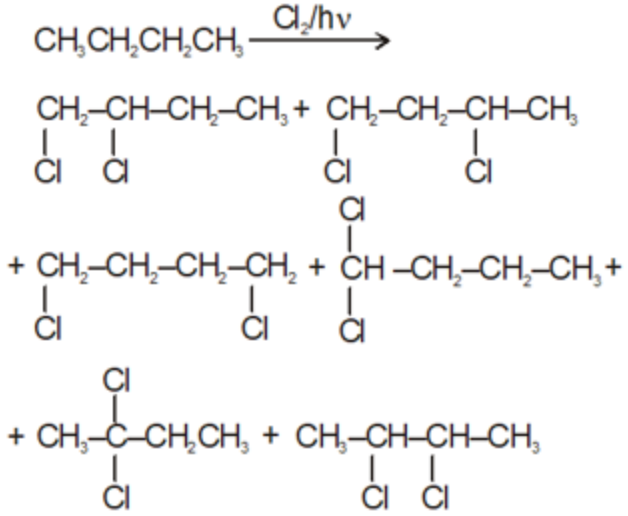
(82) Answer : (3)
Hint:
Steam distillation is used to separate substances which are steam volatile and are immiscible with water.
Solution:
Paper chromatography is a type of partition chromatography.
Glycerol can be separated from spent-lye in soap industry by using distillation under reduced pressure technique.

(83) Answer : (4)
Hint:
Activating groups like —NH_2, —OH etc. enhance rate of electrophilic aromatic substitution reactions at o, p positions.
Solution:





(84) Answer : (3)
Solution:



(85) Answer : (3)
Solution:

$$2K_2HgI_4 + NH_3 + 3KOH \rightarrow \underset{\left(\text{Basic mercury (II) amino} - \text{Iodine}\right)}{HgO \cdot Hg(NH_2)I} + 7KI + 2H_2O$$

(86) Answer : (1)
Solution:

	$CH_3COONa + HCl \longrightarrow$		$CH_3COOH + NaCl$
t = 0	1 M	1 M	0
t = t	0	0	1M

$[H^+] = \sqrt{K_a \times C}$

$pH = \frac{1}{2} pK_a = a$

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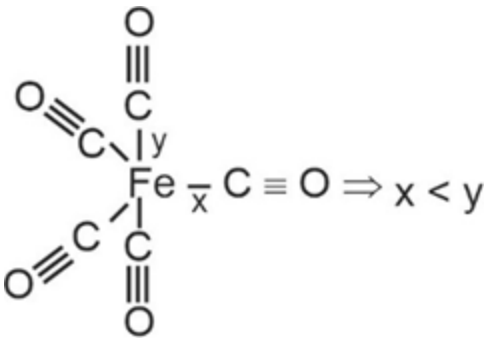
$b = pK_a + \log\frac{[CH_3COONa]}{[CH_3COOH]}$

$b = pK_a$

$a : b = \frac{pK_a}{2 \times pK_a} = 1 : 2$

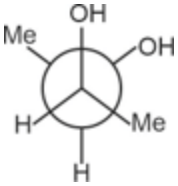
(87) Answer : (3)
Solution:

- Valence electrons in $Fe(CO)_5$ = {(n – 1)d + ns}e^- + Number of electrons from ligands
= 8 + 10 = 18
-



Due to synergic bonding C – O bond length in $Fe(CO)_5$ will be longer than C – O bond length of carbon monoxide

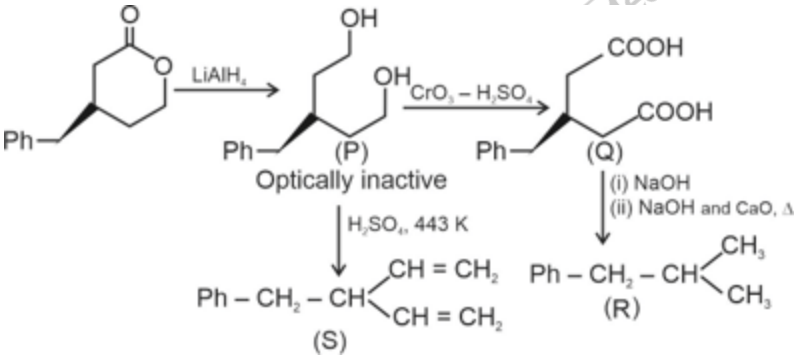
(88) Answer : (2)
Solution:



represents one of the optically active form of butane-2, 3-diol.

(89) Answer : (3)
Solution:
μ (CH_3CH_2Cl) > μ ($CH_2 = CH – Cl$)

(90) Answer : (2)
Solution:



BIOLOGY

(91) Answer : (2)
Solution:
The largest known human gene is dystrophin, with 2.4 million bases and TDF gene is the smallest gene with 14 bases.

(92) Answer : (2)
Solution:

Endothecium has α-cellulosic fibrous bands arising from inner tangential wall, which help in the dehiscence of anther due their hygroscopic nature.

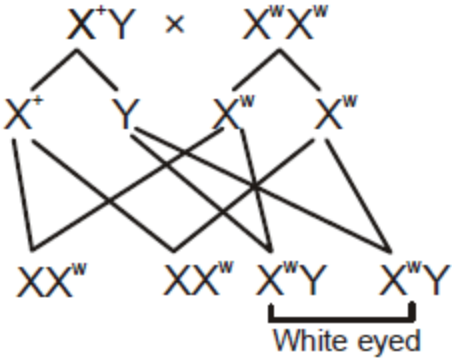
(93) **Answer :** (2)
Solution:
In RNA, every nucleotide residue has an additional –OH group present at 2' -position in the ribose.

(94) **Answer :** (4)
Solution:
Anemophilous flowers do not produce nectar.

(95) **Answer :** (4)
Solution:
Trisomy = 2n + 1. Trisomy of autosomes is presence of additional copy of autosomes.

(96) **Answer :** (3)
Solution:
Genetic variation in a population arises due to mutation as well as recombination.

(97) **Answer :** (2)
Solution:



(98) **Answer :** (3)
Solution:
Archegoniates, are group of plants having a female sexual organ in the form of an archegonium. Pteridophytes and gymnosperms are vascular archegoniates.

(99) **Answer :** (1)
Solution:
In Maize stem hypodermis is made up of sclerenchyma (Lignified, dead cells).

(100) **Answer :** (3)
Solution:
Most of the organelle duplicates in post-mitotic phase whereas centrosome, which is an organelle duplicates during the S-phase and interkinesis. Mitochondria, chloroplast and Golgi bodies duplicates in pre-mitotic (G_2) phase.

(101) **Answer :** (3)
Solution:
Primary cell wall shows growth but diminishes gradually.

(102) **Answer :** (3)
Solution:
Synaptonemal complex binds homologous chromosomes together.

(103) **Answer :** (2)
Solution:
Chromatin fibre duplication
Genetic material - 4C
Chromosome duplication
DNA replication
} In nucleus

Centriole duplication in cytoplasm.

(104) **Answer :** (3)
Solution:
Sexual spore in Basidiomycetes is basidiospore.

(105) Answer : (3)

Solution:

Slime moulds show both asexual as well as sexual reproduction.

(106) Answer : (2)

Solution:

Bioethics may be viewed as a set of standards that may be used to regulate our activities in relation to the biological world.

(107) Answer : (2)

Solution:

Pst I restriction site is present within the ampicillin resistance gene, thus it undergoes insertional inactivation if the gene of interest is ligated at *Pst* I. After the transformation with pBR322, the *E*. *coli* cells that have taken up only the plasmid will be $amp^R tet^R$ and will be able to form colonies on an agar medium that contains ampicillin or tetracycline; non-transformants will be $amp^S tet^S$.

(108) Answer : (1)

Hint:

These cells are also known as interstitial cells.

Solution:

Sertoli cells are also known as sustentacular cells. They release ABP and inhibin. Spermatozoa formation requires testosterone. Juxtaglomerular cells secrete erythropoietin.

(109) Answer : (2)

Solution:

Aldosterone acts mainly at distal tubules and stimulates reabsorption of Na^+ and water and excretion of K^+ and phosphate ions. PTH increases the Ca^{2+} levels in the blood and stimulates reabsorption of Ca^{2+} by the renal tubules.
TCT plays a significant role in maintaining the calcium balance in the body.
Oxytocin acts on smooth muscles of the body and stimulates their contraction.

(110) Answer : (1)

Solution:

Bulbourethral glands are part of male reproductive system. Each fallopian tube is 10-12 cm long and the shape of uterus is like an inverted pear.

(111) Answer : (4)

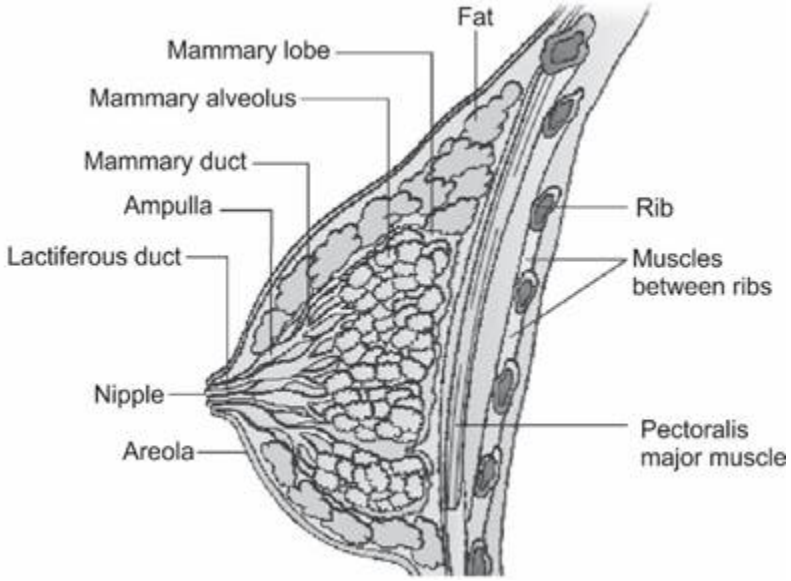
Solution:

Except for hepatitis B, genital herpes and HIV infections, other STIs are completely curable, if detected early and treated properly.

(112) Answer : (2)

Solution:

Hint: Milk eject out through lactiferous duct
Sol.:



(113) Answer : (1)

Solution:

In molluscs, such as *Sepia*, external calcareous shell is absent. In molluscs, a soft and spongy layer of skin forms a mantle over the visceral hump. The space between the hump and the mantle is called the mantle cavity in which feather-like gills are present. They have respiratory and excretory functions. The anterior head region of molluscs has sensory tentacles. Their mouth contains a file-like rasping organ for feeding, called radula.

(114) Answer : (4)
Hint:
Equal to the number of phalanges in one upper limb of humans
Solution:
Vertebrosternal ribs are also known as true ribs. First seven pairs of ribs are called true ribs. Dorsally, they are attached to the thoracic vertebrae and ventrally connected with the sternum with the help of hyaline cartilage.

(115) Answer : (3)
Solution:
In juxta-medullary nephrons, the loop of Henle extends deep into the medulla.

(116) Answer : (2)
Solution:
In *Spirogyra,* there is the predominance of chlorophyll *a* and *b*.
Eudorina is an example of colonial alga.
Complex post-fertilisation development is seen in red algae.

(117) Answer : (3)
Solution:
Bryophytes are first embryophytes. Few members of pteridophytes are heterosporous.

(118) Answer : (4)
Solution:
Sorghum is a C_4 plant, while tomato is a C_3 plant. C_4 plants show photosynthetic saturation at about 360 µl L^{-1} of CO_2 concentration, while C_3 only beyond 450 µl L^{-1}. In *Sorghum*, C_4 pathway occurs in mesophyll and bundle sheath cells of leaves. In C_4 plants, photorespiration does not occur.

(119) Answer : (3)
Solution:
During the conversion of succinyl CoA to succinic acid a molecule of ATP is synthesized.

(120) Answer : (4)
Solution:
Glycolysis and fermentation are oxygen independent reactions.

(121) Answer : (2)
Solution:
It is carboxylation step.

(122) Answer : (1)
Hint:
Cytokinin along with auxin causes callus proliferation.
Solution:
Extract of vascular tissues, yeast extract, degraded product of autoclaved herring sperm DNA and coconut milk are source of cytokinin. They along with auxin can proliferate callus.

(123) Answer : (1)
Solution:
Turgidity of plant cells helps in extension growth. Thus, plant growth and further development is intimately linked to the water status of plant.

(124) Answer : (1)
Solution:
B represents pericycle which is in the form of semilunar patches comprise of sclerenchymatous cell.
C represents cambium, meristematic tissue.
D represents metaxylem (later formed primary xylem).

(125) Answer : (2)
Solution:
Tobacco is a fumigatory plant of Solanaceae family, which shows bisexual flower with epipetalous condition of stamens.

(126) Answer : (4)
Solution:
Argemone – Parietal placentation.

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Brinjal – Valvate aestivation in calyx.
Lily – Epiphyllous condition, exemplifying adhesion of stamens.

(127) Answer : (2)
Solution:
Endocarp is hard and stony in drupe fruits.

(128) Answer : (4)
Solution:
Systematics takes into account evolutionary relationships between organisms.
The word systematics is derived from the Latin word 'Systema' which means systematic arrangement of organisms.

(129) Answer : (4)
Solution:
Apomixis produces genetically identical progeny.

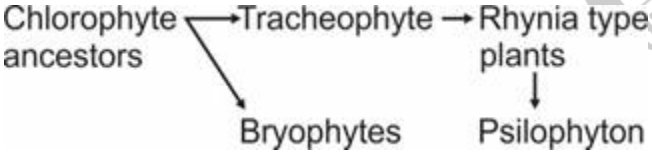
(130) Answer : (3)
Solution:
This pedigree shows the inheritance pattern of an autosomal recessive trait. Haemophilia is an X-linked recessive trait. As individual I (2) is affected, it must have received one allele from each parent and the parents are normal, which implies that both parents are heterozygous (carrier) for the disease. If the I (6) marries a female who is homozygous dominant for the given trait, then all of their offsprings will be normal. Both II (2) and II (3) can be heterozygous or homozygous for the trait.

(131) Answer : (3)
Hint:
Sarcomere is a feature of striated muscle fibres.
Solution:
Smooth/visceral muscle fibres are non-striated and uninucleated. Syncytial appearance is a property of skeletal muscle fibres. Gap junctions are present in cardiac and smooth muscle fibres.

(132) Answer : (4)
Solution:
Angina can occur in men and women of any age but it is more common among the middle aged and elderly. It occurs due to conditions that affect the blood flow.

(133) Answer : (1)
Solution:
The correct pathway for the passage of sperms in male frog is:
Testes → Vasa efferentia → Kidney → Bidder's canal → Urinogenital duct → Cloaca.

(134) Answer : (3)
Solution:



(135) Answer : (2)
Solution:
The Neanderthal man had a cranial capacity of 1400cc.

(136) Answer : (3)
Solution:
• AIDS occurs due to HIV while cancer can occur by physical, chemical or biological agents.
• Neither AIDS nor cancer spreads by mere touch or physical contact. AIDS spreads *via* body fluids and cancer is a non-communicable disease.
• Radiotherapy and immunotherapy are only used for treatment of cancer while AIDS has no cure, only prevention is the best option for it.

(137) Answer : (2)
Solution:
Bt toxin is produced by a bacterium called *Bacillus thuringiensis*.

(138) Answer : (3)
Solution:
Bronchioles and oviducts are lined by simple ciliated columnar epithelium. Simple squamous epithelium is present in alveoli of lungs.

(139) Answer : (3)

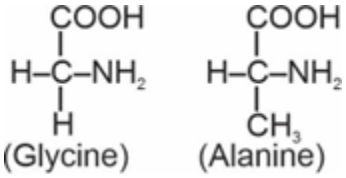
Solution:
Rib cage is ventrally formed by sternum, dorsally by vertebral column and laterally by ribs.

(140) Answer : (4)
Solution:
Use and disuse of organs is a key concept of Lamarck's theory of evolution.

(141) Answer : (2)
Solution:
By application of different methods under tissue culture, it is possible to achieve propagation of a large number of plants in very short duration.

(142) Answer : (2)
Solution:
Hymen is a part of external genitalia in females.

(143) Answer : (3)
Solution:



(144) Answer : (1)
Solution:
The testes are situated outside the abdominal cavity within a pouch called scrotum. The scrotum helps in maintaining the low temperature of the testes (2–2.5°C lower than the normal internal body temperature) necessary for spermatogenesis

(145) Answer : (4)
Solution:
Excretory organ of *Saccoglossus* is proboscis gland.
Petromyzon (Lamprey) and *Myxine* (Hag fish) are cyclostomes.

(146) Answer : (3)
Solution:
Use of disarmed retrovirus is an indirect method of gene transfer.

(147) Answer : (3)
Solution:
There is an estimated 2,00,000 varities of rice in India.

(148) Answer : (4)
Solution:
Cephalisation (formation of cephalon/brain) first of all evolved in platyhelminths. *Taenia* is unsegmented, whereas *Pheretima* exhibits true metamerism. Over two-thirds of all the named species on Earth are arthropods. Echinoderms have endoskeleton of calcareous ossicles.

(149) Answer : (2)
Solution:
Tolerance of receptors increases upon repeated intake of drugs and alcohol.

(150) Answer : (2)
Hint:
$E + S \rightleftharpoons ES \rightarrow EP \rightarrow E + P$
Solution:
The catalytic cycle of an enzyme action can be described in the following steps:
(a) The substrate binds to the active site of the enzyme.
(b) The binding of the substrate induces the enzyme to alter its shape, fitting more tightly around the substrate.
(c) The active site of the enzyme, now in close proximity of the substrate, breaks the chemical bonds of the substrate and the new enzyme product complex is formed.
(d) The enzyme releases the products of the reaction and the free enzyme is ready to bind to another molecule of the substrate.

(151) Answer : (3)
Hint:
Maximum number is equal to the number of ossicles in each ear.
Solution:
Each testicular lobule contains 1–3 highly coiled seminiferous tubules in which sperms are produced.

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(152) Answer : (4)

Solution:

Mammals have the ability to produce a concentrated urine. The Henle's loop and vasa recta play a significant role in this.

(153) Answer : (4)

Hint:

Hypothalamus lies at the base of thalamus.

Solution:

The medulla oblongata contains centres which control respiration, cardiovascular reflexes and gastric secretions.
Thermoregulatory centre is present in the hypothalamus.

(154) Answer : (3)

Hint:

Monomer of chitin is NAG.

Solution:

RNA also contains cytosine and uracil which are substituted pyrimidines. Chitin is a homopolymer of N-acetylglucosamine.

(155) Answer : (4)

Solution:

Both frogs (Presence of vocal sacs and copulatory pads in males) and cockroaches (anal style in males) show sexual dimorphism.
Frogs have a myogenic heart; cockroach has a neurogenic heart.
Frogs have haemoglobin; cockroaches do not have any respiratory pigment.
Adult frogs-Ureotelic
Cockroaches-Uricotelic

(156) Answer : (1)

Solution:

A very low level of expression of *lac* operon present in the cell all the time, so that lactose can enter the cells. The lactose then induces the operon. In the presence of an inducer, such as lactose, the repressor is inactivated by interaction with the inducer. This allows RNA polymerase access to the promoter and transcription of structural genes proceeds.

(157) Answer : (1)

Solution:

During translation, the enzyme that activates amino acid is Amino acyl-tRNA synthetase.

(158) Answer : (2)

Solution:

Antibiotics were discovered by Alexander Fleming, while working on *Staphylococcus* bacteria.

(159) Answer : (4)

Solution:

Biodiversity in Amazonian rain forest.

(1) Mammals – 427
(2) Birds – 1300
(3) Reptiles – 378
(4) Invertebrates – More than, 1,25,000

(160) Answer : (4)

Solution:

Both predation and parasitism show (+; -) type of population interaction.

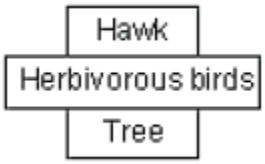
(161) Answer : (4)

Solution:

Five closely related species of warblers living on the same tree were able to avoid competition by showing behavioural differences in their foraging activities.

(162) Answer : (2)

Solution:



(163) Answer : (3)

Solution:
80S ribosomes are not found in the chloroplasts.

(164) Answer : (1)
Solution:
Recombination frequency of genes are dependent on the distance between genes.
Male grasshopper has only one sex chromosome in its somatic cell whereas female has two sex chromosomes.

(165) Answer : (3)
Solution:
The correct sequence is as follows
(d) Electrons from NADH are oxidised by NADH dehydrogenase.
(c) Ubiquinol gets oxidised with the transfer of electrons to cytochrome c via complex III.
(a) The mobile electron carrier found attached on the outer surface of inner membrane transfers the electron to complex IV.
(b) Oxygen combines with protons to form metabolic water.

(166) Answer : (4)
Solution:
Fungi show 9 + 2 arrangement in flagella. Bacterial cell wall is made of peptidoglycan.

(167) Answer : (4)
Solution:
80S ribosomes consist of 60S and 40S subunits.

(168) Answer : (3)
Hint:
The fimbriae are small bristle-like fibres in bacteria that help in attaching to the host tissue.
Solution:
Chromatophores are membranous structures containing photosynthetic pigments. Plasmid is small circular DNA that provide certain phenotypic character such as resistant to antibiotics to such bacteria. Gas vacuoles provide buoyancy.

(169) Answer : (2)
Solution:
Decomposition of detritus is an oxygen requiring process. It is faster when detritus contains water soluble sugars, proteins, nucleic acid, and it is slow if it contains lignin and chitin.

(170) Answer : (4)
Solution:
(a) There are four major causes of biodiversity losses.
(i) Habitat loss & fragmentation
(ii) Over- exploitation.
(iii) Alien species invasions.
(iv) Co-extinctions.

(171) Answer : (4)
Hint:
Response to stimulus is a feature of structural & functional units of neural system.
Solution:
Extensibility is a property of muscle fibres while neurons exhibit both excitability and conductivity.

(172) Answer : (1)
Solution:
Lactational amenorrhoea occurs when the mother breast feeds the child fully. Because any hormone is not used in natural methods of contraception, so their side effects are almost nil.

(173) Answer : (1)
Solution:
Blood is the medium of transport for O_2 and CO_2. About 97 per cent of O_2 is transported by RBCs in the blood. The remaining 3 per cent of O_2 is carried in a dissolved state through the plasma. Nearly, 20-25 per cent of CO_2 is transported by RBCs whereas 70 per cent of it is carried as bicarbonate ions. About 7 per cent of CO_2 is carried in a dissolved state through plasma.

(174) Answer : (3)
Solution:
A patch of nodal tissue is present in the right upper corner of the right atrium called the Sino-Atrial Node (SAN). Another mass of this tissue is seen in the lower left corner of the right atrium close to the atrio-ventricular septum called the Atrio-Ventricular Node (AVN). The SAN can generate the maximum number of action potentials, *i.e.*, 70-75 min^{-1}, and is responsible for

initiating and maintaining the rhythmic contractile activity of the heart. Therefore, it is called the pacemaker of human heart.

Our heart normally beats 70-75 times in a minute (average 72 beats min^{-1}).

(175) Answer : (2)

Solution:

$Na^+ – K^+$ pump transports $3Na^+$ outwards for $2K^+$ into axoplasm across resting membrane.

(176) Answer : (4)

Solution:

RNAi is the method of cellular defense in eukaryotic organisms.

(177) Answer : (3)

Solution:

ANF decreases blood pressure.

(178) Answer : (3)

Hint:

Identify the connecting link between living organisms and non-living entities.

Solution:

Interferons are glycoproteins synthesized in virus infected cells and these interferons affect neighbouring cells by synthesizing translation inhibitory proteins.

(179) Answer : (3)

Hint:

Thrombocytes are platelets.

Solution:

Leukocytes averages 6000 to 8000 mm^{-3} of blood.

(180) Answer : (3)

Hint:

Structure where ribs are attached.

Solution:

Thymus gland is a lobular structure located between lungs behind sternum on the ventral side of aorta.

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